

MAHESH PATEL

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EDUCATION

Thadomal Shahani Engineering College Mumbai University <i>B.E. in Artificial Intelligence and Data Science (CGPA: 7.78)</i>	Aug 2023 – June 2027 Mumbai
SVKM's Mithibai Junior College <i>Higher Secondary Certificate (HSC: 88.5%)</i>	Aug 2021 – March 2023 Mumbai

EXPERIENCE

Research Intern – Graph Neural Networks <i>Advisor – Dr. Himani Deshpande</i>	Dec 2025 – Jan 2026
Conducted a systematic comparative study of GCN, GAT, and hybrid architectures for node classification on benchmark citation network datasets	

PROJECTS

Municipal Grievance Routing Engine <i>Python, Web Scraping, RAG, Scikit-Learn</i>	
- Engineered an automated data pipeline to scrape, clean, and structure 350K+ unstructured records from Twitter and government portals into analysis-ready datasets - Developed a Hybrid Neural Network (CNN + BiLSTM + Attention) achieving high-accuracy multi-class grievance classification across 30+ municipal categories - Built a RAG-powered query system to surface actionable insights from structured grievance data for public policy analysis	
EduPulse – Learning Behavior Analytics & Segmentation Engine <i>Python, Pandas, Scikit-Learn, Seaborn, Plotly</i>	
- Built an end-to-end analytics pipeline to analyze user engagement patterns (session duration, drop-off points, completion rates) across learning modules - Applied clustering (K-Means, DBSCAN) to segment learners and identify content optimization opportunities	
AI-Powered Workforce Scheduler <i>FastAPI, Multi-agent CrewAI, LangChain, PostgreSQL</i>	
- Designed and implemented data models mapping complex scheduling constraints across workforce databases using PostgreSQL and SQLAlchemy - Built end-to-end automation workflows utilizing LLMs and multi-agent orchestration to optimize staff allocation decisions	

TECHNOLOGIES

Programming Languages: C, Python, Java, SQL, HTML, CSS
Data Science & ML: NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, PyTorch
Libraries/Frameworks: Node.js, Express.js, Flask, FastAPI, BeautifulSoup, Selenium, LangChain
Development Tools: Git, GitHub, Linux, AWS, Jupyter, VS Code, PyCharm, IntelliJ
Databases: Supabase/PostgreSQL, SQLAlchemy
Others: Microsoft Excel, Google Sheets, Power BI, Tableau

ACHIEVEMENTS

Augenblick Hackathon 2026 Finalist
2nd Rank at IES MCRC Hackathon 2.0 in Round 1 and Round 2 Qualifier, March 2025
Qualified GATE 2026, in Artificial Intelligence & Data Science (DA) Paper